Environmental Research 289 (2026) 123378



Contents lists available at ScienceDirect

Environmental Research

journal homepage: www.elsevier.com/locate/envres





Associations between prenatal $PM_{2.5}$ exposures and intellectual disability: Are there differential impacts based on co-occurrence of autism spectrum disorder?

Roger A. Renteria [a,*], Sara E. Grineski [a], Jacqueline Gomez [b], Deborah Bilder [c], Timothy W. Collins [b], Amanda V. Bakian [c]

[a] University of Utah, Department of Sociology, 380 S. 1530 E., Rm. 301, Salt Lake City, UT 84112, USA
[b] University of Utah, School of Environment, Society and Sustainability, 260 Central Campus Dr., Rm. 4728, Salt Lake City, UT 84112, USA
[c] University of Utah School of Medicine, Department of Psychiatry, 30 N. 1900 E., Salt Lake City, UT 84132, USA

ARTICLE INFO

Keywords:
Intellectual disability
Autism spectrum disorder
Prenatal exposure
$PM_{2.5}$
Air pollution

ABSTRACT

Prenatal fine particulate matter ($PM_{2.5}$) has been associated with intellectual disability (ID) and autism spectrum disorder (ASD), separately. Its role in co-occurring ID with ASD (ID + ASD) is less understood. To address this gap, this study uniquely disaggregates ID-only from ID + ASD to assess the associations between prenatal $PM_{2.5}$ exposure and odds of ID-only and ID + ASD across multiple exposure windows and measures. For children born in Utah, U.S.A., children with ID (ID-only and ID + ASD) were matched 1:3 with non-ID affected controls. $PM_{2.5}$ exposure was estimated across four prenatal periods (i.e., preconception, 1st, 2nd, and 3rd trimester) using long-term averages and exceedances of WHO and U.S. NAAQS 24-h thresholds and assigned to children. Associations were examined using multilevel multinomial logistic regressions. Results show that increase in long-term $PM_{2.5}$ during preconception and first trimester was associated with increased odds of ID-only and ID + ASD. Exceedances of the WHO 24-h threshold (15 $\mu g/m^3$) during preconception were linked to ID-only, while first trimester WHO 24-h exceedances were linked to both ID-only and ID + ASD. U.S. NAAQS 24-h threshold (35 $\mu g/m^3$) exceedances during preconception and first trimester were associated with ID + ASD. Results reveal that long-term $PM_{2.5}$ exposure during preconception and the first trimester elevates risk for both ID-only and ID + ASD. ID + ASD appears particularly sensitive to preconception and first trimester exceedances of the U.S. NAAQS 24-h threshold, whereas ID-only may be more sensitive to exceedances of the WHO 24-h threshold during preconception. By distinguishing these neurodevelopmental outcomes, these results underscore timing and magnitude of prenatal $PM_{2.5}$ exposure differentially influence neurodevelopmental risk. They also highlight the need for targeted public health interventions and stronger air quality regulations to protect early neurodevelopment.

1. Introduction

A growing body of scientific evidence links fine particulate matter ($PM_{2.5}$) exposure to a range of health effects across the life course (Brauer et al., 2024; World Health Organization, 2024). Exposure to both chronic and acute/peak air pollution is health-harming, especially during certain developmental windows, such as prenatal and neonatal periods (Aguilera et al., 2023).

Prenatal $PM_{2.5}$ exposure has been associated with adverse pregnancy outcomes like low birth weight, premature birth, stillbirth, miscarriage, and small size of fetus for gestational age (Bell et al., 2007; Gray et al., 2010; Gray et al., 2014, Grippo et al., 2018). Emerging research indicates prenatal $PM_{2.5}$ exposure as an important risk factor for neurodevelopmental outcomes, like intellectual disability (ID) and autism spectrum disorder (ASD) (Chun et al., 2020; Dutheil et al., 2021; Grineski et al., 2023, 2024). ID is a neurodevelopmental disability that affects intellectual and adaptive functioning, while ASD affects interactions with others, communication, and behavior. ID affects about 1 % and ASD about 3 % of the U.S. population, with their co-occurrence estimated at 39 % (Braun et al., 2015; Shaw et al., 2025).

Two studies have linked long-term $PM_{2.5}$ exposure (e.g., daily average concentrations over months or years) and exposure to

* Corresponding author.
E-mail address: roger.renteria@soc.utah.edu (R.A. Renteria).

https://doi.org/10.1016/j.envres.2025.123378
Received 24 July 2025; Received in revised form 31 October 2025; Accepted 19 November 2025
Available online 19 November 2025
0013-9351/© 2025 Published by Elsevier Inc.

exceedances of established 24-h $PM_{2.5}$ concentration thresholds during the prenatal period, specifically during the preconception and first trimester periods, to increased ID risk (Grineski et al., 2023, 2024). Grineski et al. (2023) found that higher daily average $PM_{2.5}$ exposure during the preconception period (i.e., quartiles 3–4 vs. 1) and the first trimester (i.e., quartile 4 vs. 1) was associated with an increased risk of ID. While exceedances of regulatory guidelines are less often studied, an increase in number of days exceeding the current World Health Organization (WHO) and U.S. Environmental Protection Agency (EPA) National Ambient Air Quality Standard (NAAQS) 24-h (24-h) $PM_{2.5}$ exposure thresholds ($\geq$15 μg/m^3 and $\geq$35 μg/m^3, respectively) was associated with an increased risk of ID during preconception and the first trimester (Grineski et al., 2024).

A number of studies have examined the link between prenatal $PM_{2.5}$ and ASD. Meta-analyses found an increase in ASD risk associated with long-term/average $PM_{2.5}$ exposure in the second and third trimester (Chun et al., 2020; Dutheil et al., 2021). Another found independent effects in each trimester, with the effect size being smaller in the first trimester and larger in the second and third (Liu et al., 2023). Within the literature on prenatal fine particulate matter exposure and ASD, a limited number of studies have specifically examined ASD-only (i.e., without ID) and the co-occurrence of ASD with ID (ASD + ID). In a multisite U.S. study, null associations were found between $PM_{2.5}$ exposure during the preconception period and each trimester for both ASD-only as well as ASD + ID (McGuinn et al., 2020). Similarly, no significant associations were found between full-pregnancy PM_{10} exposure and either ASD-only or ASD + ID in Stockholm, Sweden (Gong et al., 2017). To the best of our knowledge, associations between WHO or NAAQS 24-h threshold exceedances during pregnancy and ASD risk have not been examined.

Despite growing evidence connecting prenatal $PM_{2.5}$ exposure to ID and ASD separately, there is no known study that has examined how exposure to long-term $PM_{2.5}$ and exceedances of WHO and U.S. NAAQS 24-h $PM_{2.5}$ thresholds relate to their co-occurrence. While studies like Grineski et al. (2023) and (2024) focused on cases of ID with and without ASD aggregated together, the present analysis separates ID-only from ID and co-occurring ASD, allowing us to identify differential exposure-response patterns that have not previously been examined. We address the following research questions: Is prenatal $PM_{2.5}$ exposure related to greater odds of having both ID and ASD (ID + ASD) and ID-only relative to controls? Is the strength of the relationship between prenatal $PM_{2.5}$ and ID + ASD stronger than it is for ID-only? Do these relationships vary depending on long-term (i.e., trimester average) versus short-term (i.e., count of exceedance days of WHO and U.S. NAAQS 24-h thresholds during prenatal period) $PM_{2.5}$ exposures?

2. Data and methods

2.1. Study population

This study includes children identified with ID (cases) in 2010 and 2012 by the Utah Registry of Autism and Developmental Disabilities (URADD) through a retrospective population-based identification methodology developed by the Autism and Developmental Disabilities Monitoring (ADDM) Network. Case identification (children with ID-only and children with ID + ASD) focused on children born in 2002 and 2004 (ascertained at age 8) and in 2006 and 2008 (ascertained at age 4) who lived in Davis, Salt Lake, and Tooele counties (Grineski et al., 2023). Table 1 provides an overview of basic sociodemographic information for these 3 counties relative to the rest of Utah. Davis, Salt Lake, and Tooele counties together make up a large share of Utah's population, with Salt Lake County being the most populous and most diverse (32 % people of color, 13 % foreign-born). Davis and Tooele counties have similar sociodemographic profiles to the rest of Utah with younger populations, less diversity, and a lower percentage of households making <$50,000/year.

Table 1
Sociodemographic characteristics of selected Utah counties compared to the rest of the state.

	Davis County	Salt Lake County	Tooele County	Rest of Utah
Total Population	362679	1185212	70981	496233
Percent people of color	19 %	32 %	19 %	21 %
Percent Under 18 years	31 %	26 %	32 %	32 %
Percent of households making <$50,000/year	19 %	25 %	20 %	22 %
Percent foreign-born	5 %	13 %	3 %	6 %
County Rural-Urban Continuum Code	Metro 2	Metro 1	Metro 1	[a]

[a] There are 4 counties in category 2, 1 in category 3, 3 in category 4, 0 in category 5, 1 in category 6, 4 in category 7, 2 in category 8, and 7 in category 9.
Note: Total population and race data come from the 2020 census. Other data come from the 2023 5-yr American Community Survey Estimates. Rural-Urban Continuum Codes 1–3 are Metro and Codes 4–9 are Nonmetro.

Briefly, case ascertainment involved screening multiple sources and abstracting records, with ID determination based on either the most recent IQ score being $\leq$70 or expert clinician judgment. Consistent with prior studies (Bilder et al., 2013; Obi et al., 2011; Yeargin-Allsopp et al., 1992; McDonnell et al., 2019), adaptive functioning levels were not included in the ADDM ID case definition due to a risk of under-ascertainment of mild cases and children from low-income backgrounds who are more likely to be missing scores (Obi et al., 2011; Yeargin-Allsopp et al., 1992). Screening sites included, but were not limited to, state health clinics, hospitals, diagnostic centers, and providers who specialize in services for people with disabilities. More specific information on case ascertainment is provided in Grineski et al. (2023). This approach identified 296 children with ID.

Regarding ASD ascertainment, URADD deployed the ADDM methodology (Autism and Developmental Disabilities Monitoring Network Surveillance Year 2010 Principal Investigators, 2014; Christensen et al., 2016). In short, children with an ASD-related code had their health and education records screened for indictors or behaviors consistent with ASD (e.g., difficulties maintaining eye contact, sensitivity to sensory stimuli). If these indicators were present, information was abstracted from the records into a comprehensive file that was reviewed by a trained clinician. Those children whose behaviors met the DSM-IV criteria for autism were classified as having ASD. Out of the 296 cases, 137 had co-occurring ASD (ID + ASD) and 159 did not (ID-only).

The Utah Population Database (UPDB), which collates birth, medical, and death records for the state of Utah, was used to identify non-affected controls (i.e., children without a recorded diagnosis of ID, ASD, or any mental health condition). Non-affected controls (hereafter referred to as "controls") were matched 3:1 to cases based on sex, year of birth, and county of birth (n = 736).

The total sample (n = 1032) includes three analytical categories that constitute the dependent variable: controls, ID-only, and ID + ASD. Only children linked to their Utah birth certificate within the UPDB were included in the analysis, thus limiting this study to children born in Utah. Birth certificate records for cases and controls, which were accessed via UPDB, provided maternal residential address at time of birth, which were linked to exposure estimates. The project received IRB approval from the University of Utah (#00130218), as well as from the URADD Oversight Committee, the Utah Resource for Genetic and Epidemiologic Research Board (#00004589), and the Utah Department of Health and Human Services (#599).

2.2. Covariates

We included other covariates, drawn from birth certificate records for cases and controls, which are well-established risk factors for ID, to isolate the effects of $PM_{2.5}$ exposure on ID (Grineski et al., 2023, 2024;

Huang et al., 2016). These included child's race/ethnicity (i.e., non-White versus non-Hispanic White [reference category]), child's sex (i.e., male versus female [reference]), maternal age (i.e., ≤21 years, ≥35 years, and 22–34 years [reference]), maternal educational attainment (i.e., less than high school diploma, high school diploma or equivalent, some college/associate's degree, and bachelor's degree or more [reference]), maternal smoking during pregnancy (i.e., smoking versus no smoking [reference]), and conception season (i.e., spring, fall, winter, and summer [reference]). Full descriptive statistics are summarized in Grineski et al. (2024).

2.3. $PM_{2.5}$ exposure estimates

We used the U.S. EPA Downscaler product to assign daily $PM_{2.5}$ estimates to children based on maternal residential host census tract (2010 boundaries) (Grineski et al., 2023, 2024). The Downscaler $PM_{2.5}$ concentration estimates were generated via a Bayesian space-time fusion model developed by Berrocal et al. (2010). The approach fuses daily $PM_{2.5}$ monitoring data at point-level from the National Air Monitoring Stations (NAMS)/State and Local Air Monitoring Stations (SLAMS) network with gridded numerical model output from the Models-3/Community Multiscale Air Quality (CMAQ) model. Each child was assigned three $PM_{2.5}$ exposure measures including the daily average $PM_{2.5}$ exposure (μg/m^3). We examined two different magnitudes of short-term exposures: number of days with daily average $PM_{2.5}$ exceeding the WHO (≥15 μg/m^3) and U.S. NAAQS (≥35 μg/m^3) 24-h thresholds. Each of the three exposure measures were estimated across four exposure windows, including the preconception period (i.e., 12 weeks before estimated conception date), first trimester (i.e., weeks 0–12), second trimester (i.e., weeks 13–26), and third trimester (i.e., weeks 27-birth). Thus, a total of 12 $PM_{2.5}$ exposure measures were estimated per child. Descriptive statistics for the 12 $PM_{2.5}$ exposure measures are reported in Table 2 for the full sample, as well as for controls, those with ID-only and those with ID + ASD.

2.4. Statistical approach

To address missing data, we imputed 20 datasets using a regression-based method that estimates missing values for each variable. Each imputed dataset was pulled out at the 200th iteration (Enders, 2010). We applied multilevel multinomial logistic regression models to the multiply imputed data, given that our nominal dependent variable included three categories (i.e., controls, ID-only cases, and ID + ASD cases). Effects are presented as odds ratios (OR) associated with a 1 μg/m^3 increase in daily average $PM_{2.5}$ exposure or a 1-day increase exceeding the WHO or NAAQS 24-h threshold. Children were nested by birth county and birth year in the models. We estimated three model sets, each including two models based on alternative reference groups: (a) controls and (b) ID-only. Specifically, Models 1a-1b used long-term $PM_{2.5}$ variables, Models 2a-2b used the WHO short-term $PM_{2.5}$ exceedance variables, and Models 3a-3b used the NAAQS short-term exceedance variables. In each model set, the four exposure periods are considered together to avoid the bias that can stem from each exposure being added separately (Wilson et al., 2017). Correlations between the four exposure periods for each set of $PM_{2.5}$ variables were weak to moderate (i.e., 0.07 to −0.35 for average, 0 to −0.52 for WHO, and −0.04 to −0.25 for NAAQS), suggesting that entering them together was appropriate. Statistical significance was defined as $p < 0.05$ (two-tailed). Analyses were conducted using Stata v18.0.

Table 2
Descriptive statistics for dependent variable ($n = 1032$).

Variable	N (%)	% Missing
Cases and controls	1032 (100)	0
Non-ID affected controls	736 (71.3)	0
ID-only	159 (15.4)	0
ID + ASD	137 (13.3)	0

3. Results

Table 2 reports that the sample is 71.3 % controls, 15.4 % cases with ID-only, and 13.3 % cases with ID + ASD.

Table 3 compares the average daily exposure and number of exceedance days between controls, ID-only, and ID + ASD. When comparing mean exposures using ANOVA with post-hoc testing, those with ID-only and those with ID + ASD had significantly higher exposures than did controls during the preconception and first trimester; exposures for those with ID-only versus those with ID + ASD were statistically equivalent. Controls, compared to the case groups, had significantly higher average exposures during the second and third trimesters.

Table 4 presents results from multinomial logistic regression analyses examining the association between long-term and short-term $PM_{2.5}$ exposure and the three diagnostic outcomes (i.e., ID-only, ID + ASD, and controls). In Model 1a, relative to controls, a 1 μg/m^3 increase in preconception average exposure was significantly associated with increased odds of both ID-only (OR = 1.088, 95 % Confidence Interval [CI]:1.027–1.154, $p = 0.004$) and ID + ASD (OR = 1.081, CI:1.027–1.138, $p = 0.003$). Similarly, first trimester average exposure (TAE) was significantly associated with ID-only (OR = 1.077, CI:1.010–1.149, $p = 0.023$) and ID + ASD (OR = 1.086, CI:1.027–1.149, $p = 0.004$). In contrast, second and third TAE were not significantly associated with ID risk. In Model 1b, there were no significant differences in associations between $PM_{2.5}$ and the likelihood of ID + ASD versus ID-only.

In Model 2a, relative to controls, each additional day exceeding 15 μg/m^3 (WHO 24-h exceedance) was significantly associated with increased odds of ID-only (OR = 1.027, CI:1.004–1.050, $p = 0.02$); associations with ID + ASD approached significance (OR = 1.021, CI:1.000–1.043, $p = 0.053$) during the preconception period. First trimester exceedance was significantly associated with both ID-only (OR = 1.026, CI:1.002–1.052, $p = 0.037$) and ID + ASD (OR = 1.028, CI:1.005–1.052, $p = 0.016$). No significant associations were found for the second and third trimesters. In Model 2b, no significant associations were observed when comparing ID + ASD with ID-only for any exposure window.

In Model 3a, each additional day exceeding 35 μg/m^3 (U.S. NAAQS 24-h exceedance) during the preconception period was significantly associated with ID + ASD (OR = 1.045, CI:1.007–1.084, $p = 0.019$) relative to controls. The same was observed in the first trimester for ID + ASD (OR = 1.062, CI:1.022–1.103, $p = 0.002$). For ID-only, associations were also positive but not significant (preconception OR = 1.038, CI:0.994–1.084, $p = 0.095$; first trimester OR = 1.046, CI:0.998–1.095, $p = 0.059$). In contrast, exposure during the second and third trimesters was not significantly associated with either ID group. Regarding Model 3b, no statistically significant differences were observed when comparing ID + ASD with ID-only.

4. Discussion

Our results offer insights into how $PM_{2.5}$ exposures relate to ID risk with and without the co-occurrence of ASD. Much of the existing literature on $PM_{2.5}$ and neurodevelopmental outcomes has approached the question by focusing on ASD as the primary outcome (Chun et al., 2020; Dutheil et al., 2021), without considering co-occurrence with ID separately (see McGuinn et al., 2020 for an exception). While Grineski et al. (2023, 2024) examined cases of ID no matter their ASD status, this study examines the lesser studied outcome of ID first and then distinguishes between cases with or without co-occurring ASD. This allows for consideration of ID and ID + ASD as distinct outcomes, which could reveal potentially different exposure-outcome relationships that may be obscured when starting from ASD.

Table 3
$PM_{2.5}$ exposure estimates during preconception period and each trimester overall ($n = 1032$) and for controls ($n = 736$), those with ID-only ($n = 159$) and those with ID and ASD ($n = 137$), including significant differences in exposure by group.

Variables		Mean	Median	Std. Deviation	Min	Max.	ANOVA *p*
Preconception average exposure	All study participants	10.17	8.92	4.40	3.63	28.23	
	Controls (a)	9.82	8.56	4.11	3.63	28.23	$p < 0.001$
	ID-only (b)	10.93	9.80	4.59	4.91	27.51	ab, ac
	ID + ASD (c)	11.42	9.68	5.41	5.80	27.23	
Days ≥15 μg/m^3 during preconception	All study participants	12.96	9	12.79	0	52	
	Controls (a)	11.64	8	12.11	0	52	$p < 0.001$
	ID-only (b)	16.29	12	13.52	0	51	ab, ac
	ID + ASD (c)	16.23	12	14.22	0	51	
Days ≥35 μg/m^3 during preconception	All study participants	2.79	0	5.99	0	29	
	Controls (a)	2.31	0	5.47	0	29	$p < 0.001$
	ID-only (b)	3.63	0	6.48	0	28	ab, ac
	ID + ASD (c)	4.42	0	7.49	0	28	
First trimester average exposure	All study participants	10.74	9.56	4.50	4.91	27.32	
	Controls (a)	10.37	9.39	4.01	4.97	27.33	$p < 0.001$
	ID-only (b)	11.55	10.32	5.13	5.51	26.73	ab, ac
	ID + ASD (c)	11.81	9.83	5.76	4.91	27.21	
Days ≥15 μg/m^3 during first trimester	All study participants	14.84	11	13.58	0	57	
	Controls (a)	13.73	10	12.49	0	57	$p < 0.001$
	ID-only (b)	17.75	14	14.93	0	57	ab, ac
	ID + ASD (c)	17.38	11	16.50	0	54	
Days ≥35 μg/m^{3}during first trimester	All study participants	3.31	0	6.71	0	31	
	Controls (a)	2.65	0	5.94	0	31	$p < 0.001$
	ID-only (b)	4.60	0	7.62	0	29	ab, ac
	ID + ASD (c)	5.41	0	8.61	0	29	
Second trimester average exposure	All study participants	11.11	9.60	4.72	4.43	7.83	
	Controls (a)	11.51	9.83	4.93	4.43	27.83	$p < 0.001$
	ID-only (b)	10.39	9.05	3.95	6.08	26.96	ab, ac
	ID + ASD (c)	9.80	8.48	3.95	5.13	25.89	
Days ≥15 μg/m^3 during second trimester	All study participants	15.80	11	14.55	0	58	
	Controls (a)	17.18	13	14.83	0	58	$p < 0.001$
	ID-only (b)	13.33	8	13.34	0	52	ab, ac
	ID + ASD (c)	11.21	5	13.02	0	56	
Days ≥35 μg/m^3 during second trimester	All study participants	3.81	0	7.24	0	31	
	Controls (a)	4.31	0	7.71	0	31	$p < 0.01$
	ID-only (b)	2.79	0	5.86	0	29	ab, ac
	ID + ASD (c)	2.31	0	5.61	0	28	
Third trimester average exposure	All study participants	11.32	9.44	5.23	4.24	5.61	
	Controls (a)	11.61	9.64	5.44	4.47	35.61	$p < 0.05$
	ID-only (b)	10.67	9.05	4.68	5.22	30.02	
	ID + ASD (c)	10.51	9.12	4.49	4.24	26.37	
Days ≥15 μg/m^{3}during third trimester	All study participants	14.70	9	14.76	0	57	
	Controls (a)	15.70	10	15.07	0	57	$p < 0.01$
	ID-only (b)	11.89	6	13.61	0	55	ab
	ID + ASD (c)	12.56	7	13.81	0	57	
Days ≥35 μg/m^{3}during third trimester	All study participants	3.81	0	7.14	0	31	
	Controls (a)	4.23	0	7.50	0	31	$p < 0.05$
	ID-only (b)	2.75	0	6.09	0	29	ab
	ID + ASD (c)	2.76	0	5.95	0	29	

Note: The "ANOVA *p*" column reports the significance level from the ANOVA test, along with results from Tukey post-hoc comparisons where group means differ significantly. Groups are labeled as: a = control, b = ID-only, c = ID + ASD. For example, "ab" indicates a significant difference between the control group and the ID-only group.

As reflected in the bivariate analyses, average long-term $PM_{2.5}$ and exceedances of WHO and NAAQS 24-h $PM_{2.5}$ thresholds differed significantly across the groups during the prenatal period. Both ID-only and ID + ASD cases had higher mean exposures than controls during the preconception and first trimester periods, whereas controls showed higher average exposures during the second and third trimesters. However, these descriptive differences did not translate into significant effects for the control group in our multivariable models for the later prenatal periods. In the multivariable analyses, when comparing to population controls, we observed that higher long-term average $PM_{2.5}$ exposure during the preconception and first trimester periods was associated with increased odds of both ID-only and ID + ASD. In the second and third trimesters, odds ratios are weaker and not significant. These findings align with the existing study examining average $PM_{2.5}$ during the same periods and ID (with and without ASD) (Grineski et al., 2023). However, our findings do not align with the study examining ASD + ID and ASD-only, which found no significant associations between $PM_{2.5}$ and those outcomes in any trimester (McGuinn et al., 2020). Meta-analyses focused on ASD have consistently documented associations with average $PM_{2.5}$ exposure during the second and third trimester (Chun et al., 2020; Dutheil et al., 2021; Liu et al., 2023). While there are some exceptions to the second trimester window for average $PM_{2.5}$ and ASD risk (Lin et al., 2021; Mamun et al., 2025), there may be differences in the timing of $PM_{2.5}$ exposure for risk of ID + ASD vs. ASD-only that should be examined in future studies.

While ID + ASD aligns with ID-only in terms of critical exposure windows for long-term average $PM_{2.5}$, findings from the analyses of two different thresholds of short-term exposure suggest the possibility that ID + ASD has different critical prenatal exposure windows than ID-only and ID more generally. Note that we cannot compare directly to ASD due to the lack of studies using these thresholds. Regarding WHO (≥15 μg/m^3) 24-h exceedances, additional exceedance days in the first trimester are associated with both ID + ASD and ID-only. However, during the preconception period, an additional exceedance day is significantly

Table 4
Pooled multinomial logistic results for model 1a (reference category: control) and model 1b (reference: ID-only) considering average $PM_{2.5}$, model 2a (reference: control) and model 2b (reference: ID-only) considering daily exceedances of WHO 24-h $PM_{2.5}$ guidelines, and model 3a (reference: control) and model 3b (reference: ID-only) considering daily exceedances of 24-h $PM_{2.5}$ US NAAQS guidelines ($n = 1032$).

Variables	Model 1a (Ref: Non-ID affected control)						Model 1b (Ref: ID-only)					
	ID-only			ID + ASD			Control			ID + ASD		
	OR	95 % CI	*p*	OR	95 % CI	*p*	OR	95 % CI	*p*	OR	95 % CI	*p*
Preconception average exposure	1.088	1.027–1.154	0.004**	1.081	1.027–1.138	0.003**	0.923	0.874–0.974	0.004**	1.007	0.947–1.071	0.819
First TAE	1.077	1.010–1.149	0.023*	1.086	1.027–1.149	0.004**	0.934	0.881–0.991	0.024*	1.037	0.974–1.103	0.258
Second TAE	1.020	0.949–1.096	0.590	0.937	0.872–1.007	0.077	0.990	0.926–1.058	0.770	0.946	0.869–1.030	0.198
Third TAE	1.056	0.987–1.131	0.117	1.007	0.945–1.073	0.824	0.954	0.896–1.016	0.140	0.976	0.906–1.053	0.534
	Model 2a (Ref: Non-ID affected control)						Model 2b (Ref: ID-only)					
	ID-only			ID + ASD			Control			ID + ASD		
	OR	95 % CI	*p*	OR	95 % CI	*p*	OR	95 % CI	*p*	OR	95 % CI	*p*
Days ≥15 μg/m^3 during preconception	1.027	1.004–1.050	0.023*	1.021	1.000–1.043	0.053	0.975	0.954–0.997	0.023*	0.999	0.974–1.024	0.919
Days ≥15 μg/m^3 during first trimester	1.026	1.002–1.052	0.037*	1.028	1.005–1.052	0.016*	0.975	0.954–0.998	0.030*	1.010	0.984–1.035	0.456
Days ≥15 μg/m^3 during second trimester	1.004	0.976–1.032	0.793	0.976	0.950–1.003	0.077	0.998	0.973–1.024	0.897	0.980	0.949–1.012	0.216
Days ≥15 μg/m^3 during third trimester	1.009	0.982–1.036	0.531	0.996	0.972–1.021	0.761	0.993	0.969–1.018	0.601	0.995	0.966–1.025	0.746
	Model 3a (Ref: Non-ID affected control)						Model 3b (Ref: ID-only)					
	ID-only			ID + ASD			Control			ID + ASD		
	OR	95 % CI	*p*	OR	95 % CI	*p*	OR	95 % CI	*p*	OR	95 % CI	*p*
Days ≥35 μg/m^3 during preconception	1.038	0.994–1.084	0.095	1.045	1.007–1.084	0.019*	0.969	0.931–1.007	0.112	1.019	0.976–1.064	0.386
Days ≥35 μg/m^3 during first trimester	1.046	0.998–1.095	0.059	1.062	1.022–1.103	0.002**	0.962	0.923–1.002	0.059	1.035	0.994–1.078	0.097
Days ≥35 μg/m^3 during second trimester	1.004	0.955–1.055	0.884	0.972	0.927–1.019	0.239	1.003	0.960–1.049	0.889	0.987	0.935–1.042	0.636
Days ≥35 μg/m^3 during third trimester	1.019	0.965–1.075	0.506	1.000	0.955–1.048	0.984	0.988	0.942–1.037	0.629	1.002	0.950–1.057	0.929

Note: Pooled results from 20 multiple imputed datasets are presented. TAE = trimester average exposure to $PM_{2.5}$. Models adjust for child's race/ethnicity (i.e., BIPOC [Black, Indigenous and People of Color] vs. non-Latina/x/o/e White [reference]), child sex (i.e., male vs. female), maternal age (i.e., ≤21, ≥35, and 22–34 [reference]), any maternal prenatal smoking (i.e., yes vs. no), and maternal education (i.e., four categories of mother's educational attainment), and conception season (i.e., December–February, March–May, June–August (reference), and September–November). Models also account for clustering based on child's birth year and maternal residential county. Significance levels: $^{**}p < 0.01$, $^{*}p < 0.05$.

associated with ID-only, but not significantly associated with ID + ASD. At the higher 24-h exceedance threshold of the U.S. NAAQS (≥35 μg/m^3), additional exceedance days during preconception period and first trimester were significantly associated with ID + ASD, but not ID-only.

When looking at ID independent of ASD status, additional exceedances at both thresholds during preconception and the first trimester were significantly associated with odds of ID (Grineski et al., 2024). When aggregated, these subtle differences in risk profiles for ID + ASD and ID-only are masked. By separating them here, we see the possibility that ID + ASD may be more sensitive to short-term, high concentration (≥35 μg/m^3) $PM_{2.5}$ events in preconception and early pregnancy than ID-only, while ID-only might be more sensitive to short-term but lower-concentration (≥15 μg/m^3) $PM_{2.5}$ events during preconception. This raises the possibility that exposure magnitude may be differently related to ID-only and ID + ASD (and potentially ASD-only, although there is no research to date on this outcome). Future research is needed to examine this.

Nonetheless, these patterns of early exposures suggest that prenatal exposure to $PM_{2.5}$ may affect neurodevelopment through several interrelated biological mechanisms affecting fetal brain development such as oxidative stress and systemic inflammation (Calderón-Garcidueñas et al., 2008, 2016; Chiu et al., 2016). These processes are particularly consequential during sensitive windows such as the preconception and early gestational periods when rapid cell proliferation and neural tube formation occur (Zhang et al., 2020), which may help explain the significant associations observed for these early exposure windows.

4.1. Limitations

While being the first study to our knowledge to examine associations of long-term and short-term $PM_{2.5}$ exposures with ID-only and ID + ASD, this study has several limitations. These include the possibility of exposure misclassification. We assumed each child was born on the 15th of the month because data on only the month and year of birth were available, implying that our gestational window estimates exhibit >85 % accuracy in alignment with the child's actual temporal windows. Also, we cannot confirm whether the mother moved during the preconception and gestational period. As each of our exposure estimates are available only at the census tract level and we lack of biomonitoring data, there is within-tract variability in exposure that we are unable to capture. This means our $PM_{2.5}$ estimates do not completely capture individual-level exposures. However, the Downscaler product provides high-resolution daily concentration estimates validated against other $PM_{2.5}$ exposure models and has been used in a variety of epidemiologic studies (Berrocal et al., 2020; Son et al., 2020). Another limitation is that we do not have address information on where the children lived after they were born, which limits our ability to assess postnatal exposures. While we focused on $PM_{2.5}$ as our air pollutant of analysis, other studies have examined the exposure to multiple air pollutants and their associations with neurodevelopment outcomes (e.g., Hashemi et al., 2025). Future studies should expand to include other pollutants as well as their cumulative, distinct, and synergistic effects on neurodevelopmental outcomes.

Another set of limitations relates to our ID data. It is possible that we missed children with ID during the ascertainment process, although this would bias our findings towards the null. We could not examine ASD-only as another outcome, given that our data set includes only those with ID. We lack information on genetic predisposition or family history of ID or ASD. Nonetheless, our findings capture potential environmental contributions within broader multifactorial influences on ID and ASD. We aggregated Black, Asian, American Indian, Pacific Islander and multi-racial children to one racial/ethnic minority group due to small counts. As our dataset pertains only to Utah, the sample is majority White, which may limit generalizability.

5. Conclusion

Although we did not examine ASD-only, findings from ASD- and ID-focused studies in concert with our results suggest possible divergence in critical exposure windows and in the importance of the magnitude of short-term exposures across ID-only, ID + ASD, and ASD-only groups. Future research with larger sample sizes and more diverse populations should explicitly examine ID-only, ASD-only, and ID + ASD as distinct outcomes to test these emerging questions raised by this initial analysis. Our findings add to evidence that $PM_{2.5}$ exposure during the preconception period and first trimester may be associated with risk for ID. Exposures during these windows could be risky due to the vulnerability of maturing gametes to epigenetic changes from $PM_{2.5}$ exposure (Marcho et al., 2020), the quality of sperm and egg, the implantation process (Castro et al., 2013), and rapid neurological and organ development (Kwon and Kim, 2017). These windows could be targeted for interventions aimed at reducing maternal exposure to $PM_{2.5}$. Public health interventions should take a multi-faceted approach by improving maternal care, expanding access to health protective technologies (e.g., residential air filtration), and advocating for policy changes in environmental governance and regulation to improve air quality. Investing in cleaner air is not only about reducing immediate health risks but also about safeguarding future generations' cognitive health and broader human development.

CRediT authorship contribution statement

Roger A. Renteria: Writing – original draft, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Sara E. Grineski:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jacqueline Gomez:** Writing – original draft, Data curation. **Deborah Bilder:** Writing – review & editing. **Timothy W. Collins:** Writing – review & editing, Funding acquisition. **Amanda V. Bakian:** Writing – review & editing, Funding acquisition.

Data sharing

The authors cannot make the data publicly available nor are they permitted to share the data with other researchers, due to their sensitive nature. Researchers wishing to obtain the data can apply to access it directly via the Utah Resource for Genetic and Epidemiologic Research, the Utah Department of Health and Human Services, and the Utah Registry of Autism and Developmental Disabilities Oversight Committee.

Funding

This article is based on work supported by the National Institute of Environmental Health Sciences under award R25 ES031497.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sara E. Grineski reports financial support was provided by National Institutes of Health. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We appreciate the unique collaboration provided across the University of Utah, Intermountain Healthcare, Utah Registry of Autism and Developmental Disabilities, Utah Department of Health and Human Services, and the Pedigree and Population Resource (funded by the Huntsman and Intermountain Healthcare Cancer Foundation). We also acknowledge the URADD Oversight Committee, Andrew Hammer, Jared DeWitt, and Alison Fraser for their help with this project.

Data availability

The authors do not have permission to share data.

References

Aguilera, J., Konvinse, K., Lee, A., Maecker, H., Prunicki, M., Mahalingaiah, S., et al., 2023. Air pollution and pregnancy. Semin. Perinatol. 47 (8), 151838.
Autism and Developmental Disabilities Monitoring Network Surveillance Year 2010 Principal Investigators, 2014. Prevalence of autism spectrum disorder among children aged 8 years — autism and developmental disabilities monitoring network, 11 sites, United States, 2010. Morb. Mortal. Wkly. Rep. - Surveillance Summ. 63 (2), 1–21.
Bell, M.L., Ebisu, K., Belanger, K., 2007. Ambient air pollution and low birth weight in Connecticut and Massachusetts. Environ. Health Perspect. 115 (8), 1118–1124.
Berrocal, V.J., Gelfand, A.E., Holland, D.M., 2010. A spatio-temporal downscaler for output from numerical models. J. Agric. Biol. Environ. Stat. 15 (2), 176–197.
Berrocal, V.J., Guan, Y., Muyskens, A., Wang, H., Reich, B.J., Mulholland, J.A., Chang, H. H., 2020. A comparison of statistical and machine learning methods for creating national daily maps of ambient $PM_{2.5}$ concentration. Atmos. Environ. 222, 117130.
Bilder, D.A., Pinborough-Zimmerman, J., Bakian, A.V., Miller, J.S., Dorius, J.T., Nangle, B., McMahon, W.M., 2013. Prenatal and perinatal factors associated with intellectual disability. Am. J. Intellect. Dev. Disabil. 118 (2), 156–176.
Brauer, M., Roth, G.A., Aravkin, A.Y., Zheng, P., Abate, K.H., Abate, Y.H., et al., 2024. Global burden and strength of evidence for 88 risk factors in 204 countries and 811 subnational locations, 1990–2021: a systematic analysis for the global burden of disease study 2021. Lancet 403 (10440), 2162–2203.
Braun, K.V.N., Christensen, D., Doernberg, N., Schieve, L., Rice, C., Wiggins, L., et al., 2015. Trends in the prevalence of autism spectrum disorder, cerebral palsy, hearing loss, intellectual disability, and vision impairment, metropolitan Atlanta, 1991–2010. PLoS One 10 (4), e0124120.
Calderón-Garcidueñas, L., Leray, E., Heydarpour, P., Torres-Jardón, R., Reis, J., 2016. Air pollution, a rising environmental risk factor for cognition, neuroinflammation and neurodegeneration: the clinical impact on children and beyond. Rev. Neurol. 172 (1), 69–80.
Calderón-Garcidueñas, L., Solt, A.C., Henríquez-Roldán, C., Torres-Jardón, R., Nuse, B., Herritt, L., et al., 2008. Long-term air pollution exposure is associated with neuroinflammation, an altered innate immune response, disruption of the blood-brain barrier, ultrafine particulate deposition, and accumulation of amyloid β-42 and α-synuclein in children and young adults. Toxicol. Pathol. 36 (2), 289–310.
Castro, K.R., Ribeiro-Junior, R., Peres, M., Saldiva, P., Matsuda, M., Veras, M., 2013. Murine uterine receptivity markers are affected by particulate air pollution in a dose response manner. Placenta 34 (Suppl. A), A10.
Chiu, Y.H.M., Hsu, H.H.L., Coull, B.A., Bellinger, D.C., Kloog, I., Schwartz, J., et al., 2016. Prenatal particulate air pollution and neurodevelopment in urban children: examining sensitive windows and sex-specific associations. Environ. Int. 87, 56–65.
Christensen, D.L., Baio, J., Braun, K.V.N., Bilder, D., Charles, J., et al., 2016. Prevalence and characteristics of autism spectrum disorder among children aged 8 years — autism and developmental disabilities monitoring network, 11 sites, United States, 2012. Morb. Mortal. Wkly. Rep. - Surveillance Summ. 65 (3), 1–23.
Chun, H., Leung, C., Wen, S.W., McDonald, J., Shin, H.H., 2020. Maternal exposure to air pollution and risk of autism in children: a systematic review and meta-analysis. Environ. Pollut. 256, 113307.
Dutheil, F., Comptour, A., Morlon, R., Mermillod, M., Pereira, B., Baker, J.S., et al., 2021. Autism spectrum disorder and air pollution: a systematic review and meta-analysis. Environ. Pollut. 278, 116856.
Enders, C., 2010. Applied Missing Data Analysis. Guilford Press.
Gong, T., Dalman, C., Wicks, S., Dal, H., Magnusson, C., Lundholm, C., et al., 2017. Perinatal exposure to traffic-related air pollution and autism spectrum disorders. Environ. Health Perspect. 125 (1), 119–126.
Gray, S.C., Edwards, S.E., Miranda, M.L., 2010. Assessing exposure metrics for PM and birth weight models. J. Expo. Sci. Environ. Epidemiol. 20 (5), 469–477.

Gray, S.C., Edwards, S.E., Schultz, B.D., Miranda, M.L., 2014. Assessing the impact of race, social factors and air pollution on birth outcomes: a population-based study. Environ. Health 13, 1–8.

Grineski, S.E., Alexander, C., Renteria, R., Collins, T.W., Bilder, D., VanDerslice, J., et al., 2023. Trimester-specific ambient PM2.5 exposures and risk of intellectual disability in Utah. Environ. Res. 218, 115009.

Grineski, S.E., Renteria, R.A., Collins, T.W., Bakian, A.V., Bilder, D., VanDerslice, J.A., et al., 2024. PM2.5 threshold exceedances during the prenatal period and risk of intellectual disability. J. Expo. Sci. Environ. Epidemiol. https://doi.org/10.1038/s41370-024-00647-0.

Grippo, A., Zhang, J., Chu, L., Guo, Y., Qiao, L., Zhang, J., et al., 2018. Air pollution exposure during pregnancy and spontaneous abortion and stillbirth. Rev. Environ. Health 33 (3), 247–264.

Hashemi, F., Hoepner, L., Hashemi, H., Hoseini, M., Omeragić, E., Hamidinejad, F.S., et al., 2025. Prenatal exposure to benzene, toluene, ethylbenzene, and xylene (BTEX) and the risk of autism spectrum disorders in children aged 16 to 37 months: a prospective cohort study. Environ. Pollut., 126700

Huang, J., Zhu, T., Qu, Y., Mu, D., 2016. Prenatal, perinatal and neonatal risk factors for intellectual disability: a systemic review and meta-analysis. PLoS One 11 (4), e0153655.

Kwon, E.J., Kim, Y.J., 2017. What is fetal programming?: a lifetime health is under the control of in utero health. Obstetrics & Gynecology Science 60 (6), 506.

Lin, C.K., Chang, Y.T., Lee, F.S., Chen, S.T., Christiani, D., 2021. Association between exposure to ambient particulate matters and risks of autism spectrum disorder in children: a systematic review and exposure-response meta-analysis. Environ. Res. Lett. 16, 063003.

Liu, H., Ding, L., Qu, G., Guo, X., Liang, M., Ma, S., et al., 2023. Particulate matter exposure during pregnancy and infancy and risks of autism spectrum disorder in children: a systematic review and meta-analysis. Sci. Total Environ. 855, 158830.

Mamun, M.A., Al-Mamun, F., Roy, N., Raquib, A., Kaggwa, M.M., Almerab, M.M., et al., 2025. Preconception and gestational versus postnatal exposure to air pollutants and risk of autism spectrum disorder: a systematic review and meta-analysis. Int. Arch. Occup. Environ. Health 98 (1), 33–57.

Marcho, C., Oluwayiose, O.A., Pilsner, J.R., 2020. The preconception environment and sperm epigenetics. Andrology 8 (4), 924.

McDonnell, C.G., Boan, A.D., Bradley, C.C., Seay, K.D., Charles, J.M., Carpenter, L.A., 2019. Child maltreatment in autism spectrum disorder and intellectual disability: results from a population-based sample. JCPP (J. Child Psychol. Psychiatry) 60 (5), 576–584.

McGuinn, L.A., Windham, G.C., Kalkbrenner, A.E., Bradley, C., Di, Q., Croen, L.A., et al., 2020. Early life exposure to air pollution and autism spectrum disorder: findings from a multisite case–control study. Epidemiology 31 (1), 103–114.

Obi, O., Van Naarden Braun, K., Baio, J., Drews-Botsch, C., Devine, O., Yeargin-Allsopp, M., 2011. Effect of incorporating adaptive functioning scores on the prevalence of intellectual disability. Am. J. Intellect. Dev. Disabil. 116 (5), 360–370.

Shaw, K.A., Williams, S., Patrick, M.E., Valencia-Prado, M., Durkin, M.S., Howerton, E. M., et al., 2025. Prevalence and early identification of autism spectrum disorder among children aged 4 and 8 years—Autism and developmental disabilities monitoring network, 16 sites, United States, 2022. Morb. Mortal. Wkly. Rep. - Surveillance Summ. 74 (2), 1–22. https://doi.org/10.15585/mmwr.ss7402a1.

Son, J.Y., Lane, K.J., Miranda, M.L., Bell, M.L., 2020. Health disparities attributable to air pollutant exposure in North Carolina: influence of residential environmental and social factors. Health Place 62, 102287.

Wilson, A., Chiu, Y.H.M., Hsu, H.H.L., Wright, R.O., Wright, R.J., Coull, B.A., 2017. Potential for bias when estimating critical windows for air pollution in children's health. Am. J. Epidemiol. 186 (11), 1281–1289.

World Health Organization, 2024. Health consequences of air pollution. https://www.who.int/news/item/25-06-2024-what-are-health-consequences-of-air-pollution-on-populations.

Yeargin-Allsopp, M., Oakley, G.P., Murphy, C.C., Sikes, R.K., 1992. A multiple-source method for studying the prevalence of developmental disabilities in children: the metropolitan Atlanta developmental disabilities study. Pediatrics 89 (4), 624–630.

Zhang, J.Y., Wu, Q.J., Huang, Y.H., Li, J., Liu, S., Chen, Y.L., et al., 2020. Association between maternal exposure to ambient PM10 and neural tube defects: a case-control study in Liaoning Province, China. Int. J. Hyg Environ. Health 225, 113453.